



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short Questions.

(6x5=30)

1. Show that

$$\int_0^{\infty} x^{2s} e^{-ax^2} dx = \frac{\Gamma(s + \frac{1}{2})}{2a^{s+1/2}}.$$

2. Show that the Bessel functions $J_n(x)$ can also be obtained from the following Laurent series

$$g(x, t) = e^{x(t-\frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x) t^n.$$

3. Find the Fourier transform of the normalised Gaussian distribution

$$f(t) = \frac{1}{a\sqrt{2\pi}} \exp\left(-\frac{t^2}{2a^2}\right), \quad -\infty < t < \infty.$$

4. Find the Fourier series representation of

$$f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ x, & 0 \leq x < \pi. \end{cases}$$

5. Prove that

$$\left(2x - \frac{d}{dx}\right)^n 1 = H_n(x).$$

Hint: Use mathematical induction.

6. Show that

$$\Gamma\left(\frac{1}{2} - n\right) \Gamma\left(\frac{1}{2} + n\right) = (-1)^n \pi,$$

where n is an integer.

Q.2. Answer the following Questions.

(5x6=30)

1. Find the Green's function for the operator

$$Ly(x) = \frac{d^2y}{dx^2} + y, \quad y(0) = 0, \quad y'(1) = 0.$$

2. Use Fourier transformation to solve the wave equation,

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2},$$

where v is the phase velocity of the wave propagation. We take as initial conditions

$$y(x, 0) = f(x), \quad \left. \frac{\partial y(x, t)}{\partial t} \right|_{t=0} = 0,$$

where $f(x)$ is assumed localized, meaning that $\lim_{x \rightarrow \pm\infty} f(x) = 0$.

3. From

$$J_\nu(x) = \frac{1}{2\pi i} \left(\frac{x}{2}\right)^\nu \int t^{-\nu-1} e^{t-x^2/4t} dt,$$

derive the recurrence relation

$$\frac{d}{dx} J_\nu(x) = \frac{\nu}{x} J_\nu(x) - J_{\nu+1}(x).$$

4. The functions $u_1(x)$ and $u_2(x)$ are eigenfunctions of the same Hermitian operator but for distinct eigenvalues λ_1 and λ_2 . Prove that $u_1(x)$ and $u_2(x)$ are linearly independent.

5. A function $f(x)$ is expanded in a Legendre series $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$. Show that

$$\int_{-1}^{+1} |f(x)|^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}.$$

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Q#2:

Short Question:-

show that

$$g(x, t) = e^{x/2} (t^{-1/2}) = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

Solution:-

consider

$$e^{x/2} (t^{-1/2}) = e^{\frac{x}{2}} \cdot e^{-\frac{x}{2t}}$$

$$= \left(1 + \frac{x}{2}t + \frac{x^2}{2!4}t^2 + \frac{x^3}{3!8}t^3 + \dots \right) \times \left(1 - \frac{x}{2}t^{-1} + \frac{x^2}{2!4}t^{-2} - \frac{x^3}{3!8}t^{-3} + \dots \right)$$

$$= \dots t^{-2} \left(\frac{x^2}{2!4} - \frac{x^4}{2 \cdot 3!8} + \dots \right) + t^{-1} (-1) \left(\frac{x}{2} - \frac{x^3}{4 \cdot 2! \cdot 2!} + \dots \right)$$

$$+ t^0 \left(1 - \frac{x^2}{2 \cdot 2} + \dots \right) + t \left(\frac{x}{2} - \frac{x^3}{4 \cdot 2} + \dots \right) + \dots$$

putting the value of x and

we will get

$$= \dots + t^{-2} J_2(x) + t^{-1} (-1) J_1(x) + t^0 J_0(x) + t J_1(x) + \dots$$

$$t^{-2} (-1)^2 J_2(x) + \dots + \dots$$

$$J_{-n}(x) = (-1)^n J_n(x)$$

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$$= \dots + \bar{f}_{-2}(x)t^{-2} + \bar{f}_{-1}(x)t^{-1} + t^0 \bar{f}_0(x) + t^1 \bar{f}_1(x) + \dots$$

$$= \sum_{n=-\infty}^{\infty} \bar{f}_n(x) t^n$$

proved:

Q#4:

Find Fourier series

$$f(x) = \begin{cases} 0 & -\pi < x \leq 0 \\ x & 0 \leq x < \pi \end{cases}$$

solution:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{where } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \quad \text{and}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^0 0 dx + \frac{1}{\pi} \int_0^{\pi} x dx \Rightarrow \frac{1}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi}$$

$$= \frac{1}{2\pi} [\pi^2 - 0] = \frac{1}{2\pi} (\pi^2) = \frac{\pi}{2}$$

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$$b_n = \frac{1}{\pi} \int_{-\pi}^0 0 dx + \frac{1}{\pi} \int_0^{\pi} x \sin nx dx$$

$$b_n = \left(\frac{x \cos nx}{n} \right)_0^{\pi} - \int_0^{\pi} \cos nx \cdot 1 dx$$

$$b_n = \frac{x}{n\pi} \{(-1)^n + 1\} = \frac{x}{n\pi} (-1)^{n+1}$$

Hence required Fourier series

$$f(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x}{n\pi} \sin nx$$

Q#6

show that

$$\Gamma\left(\frac{1}{2}-n\right) \Gamma\left(\frac{1}{2}+n\right) = (-1)^n \pi$$

solution:-

$$\Gamma\left(\frac{1}{2}-n\right) \Gamma\left(\frac{1}{2}+n\right) = (-1)^n \pi$$

by Euler reflection Formula

$$\Gamma n \Gamma n+1 = \frac{\pi}{\sin(\pi n)}$$

Replace n by $\frac{1}{2}-n$

$$\Gamma\left(\frac{1}{2}-n\right) \Gamma\left(1-\frac{1}{2}+n\right) = \frac{\pi}{\sin\left(\left(\frac{1}{2}-n\right)\pi\right)}$$

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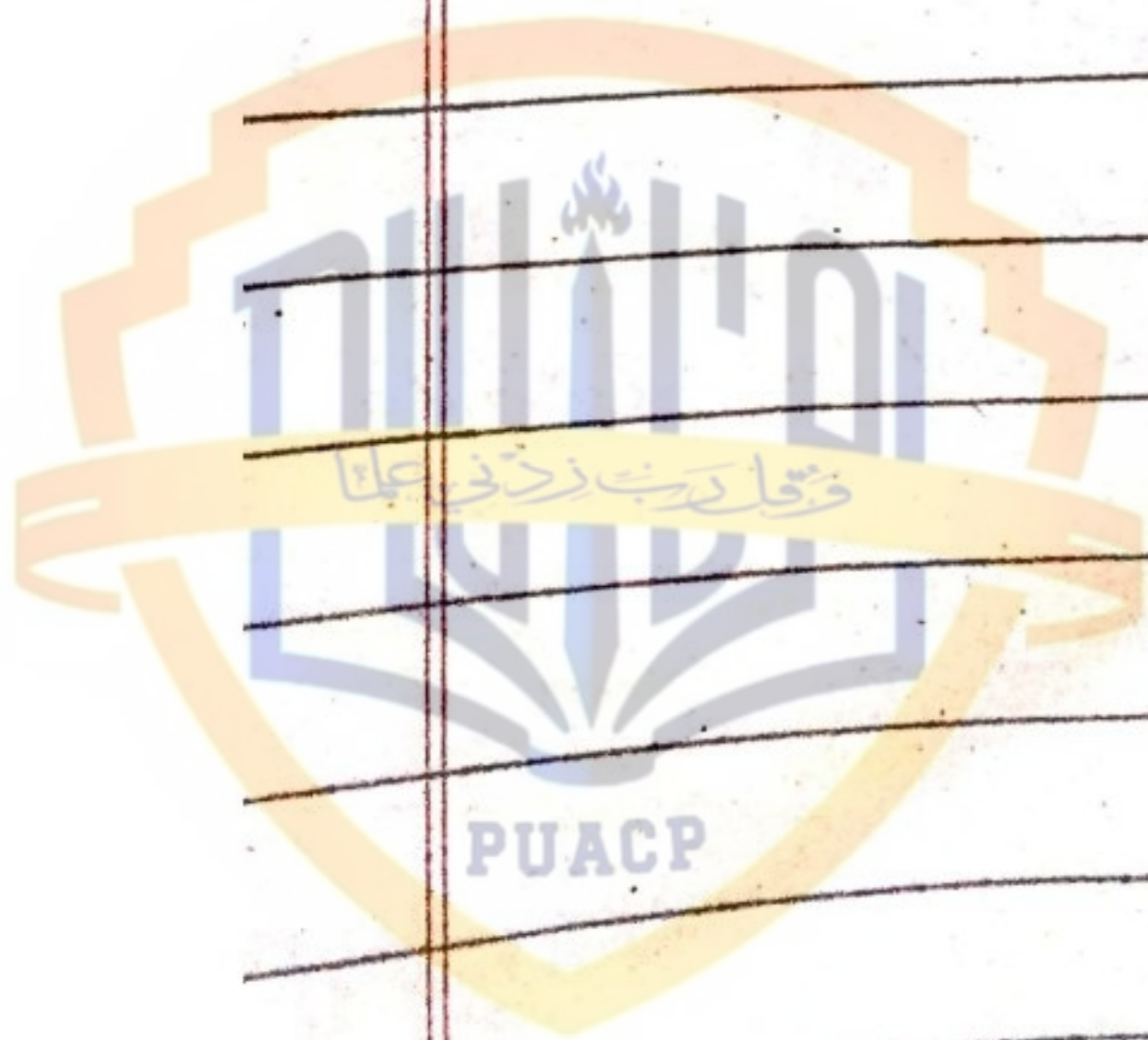
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$$\sqrt{\frac{1}{2}-n} \sqrt{\frac{1}{2}+n} = \frac{\pi}{\cos(-n\pi)} = \frac{\pi}{\cos n\pi}$$

so,

$$\sqrt{\frac{1}{2}-n} \sqrt{\frac{1}{2}+n} = \frac{\pi}{(-1)^n} \rightarrow (-1)^n \pi$$

proved



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Long Question:-

Q#2:

$$\frac{\partial^2 v}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 v}{\partial t^2}$$

where v is the phase velocity of the wave:

initial condition:- $v(x, 0) = f(x)$

$$, \frac{\partial v(x, t)}{\partial t} = 0$$

Solution:-

We solve this eq by boundary condition:

$$(i) v(x, 0) = f(x) , \frac{\partial v(x, t)}{\partial t} = 0$$

put $v(x, t) = X(x) T(t)$ in eq

$$\frac{\partial^2 v}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 v}{\partial t^2}$$

$$\frac{\partial^2 XT}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 XT}{\partial t^2} \Rightarrow T \cdot \frac{d^2 X}{dx^2} = \frac{1}{v^2} X \frac{d^2 T}{dt^2}$$

$$= \frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{T} \frac{d^2 T}{dt^2}$$

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Left side is function x only and

right side function t only

so eq is satisfied

if both side are equal

Then General solution

$$X(x) = A \cos mx + B \sin mx$$

$$T(t) = C \cos mt + D \sin mt$$

$$v(0, t) = 0 \Rightarrow X(0) = 0 \Rightarrow A = 0$$

$$v(a, t) = 0 \Rightarrow X(a) = 0 \Rightarrow B \sin mx = 0$$

$$\because B \neq 0 \quad \text{so} \quad \sin ma = 0 \Rightarrow ma = n\pi$$

$$\Rightarrow m = \frac{n\pi}{a} \quad ; \quad n = 1, 2, 3, \dots$$

Hence

$$X(x) = B_n \sin\left(\frac{n\pi}{a}x\right)$$

and

$$T(t) = C_n \cos\left(\frac{n\pi}{a}t\right) + D_n \sin\left(\frac{n\pi}{a}t\right)$$

if $v(x, t)$

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$$V(x, t) = B_n \sin\left(\frac{n\pi}{a}\right)x \left\{ C_n \cos\left(\frac{n\pi}{a}\right)ct + D_n \sin\left(\frac{n\pi}{a}\right)ct \right\}$$

$$= \sin\left(\frac{n\pi}{a}\right)x \left\{ C'_n \cos\left(\frac{n\pi}{a}\right)ct + D'_n \sin\left(\frac{n\pi}{a}\right)ct \right\}$$

So general solution is
linear combination of all
solution.

$$v(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{a}\right)x \left\{ C'_n \cos\left(\frac{n\pi}{a}\right)ct + D'_n \sin\left(\frac{n\pi}{a}\right)ct \right\}$$

Q#5:

A function $f(x)$ is expanded
in Legendre series.

$$\int_{-1}^1 f(x) dx = \sum_{n=0}^{\infty} \frac{2a_n}{2n+1}$$

where $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$

Solution:-

consider The function

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$$f(x) f(x) = \sum_{n=0}^{\infty} a_n P_n(x) \cdot \sum_{n=0}^{\infty} a_n P_n(x)$$

$$f(x)^2 = \sum_{n=0}^{\infty} a_n^2 P_n^2(x)$$

integrate w.r.t x

From -1 to 1

$$\int_{-1}^1 f(x)^2 dx = \sum_{n=0}^{\infty} a_n^2 \int_{-1}^1 P_n^2(x) dx$$

$$= \sum_{n=0}^{\infty} a_n^2 \frac{2}{2n+1}$$

$$\int_{-1}^1 f(x)^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}$$

proved:



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Q#3:-

Derive Recurrence Relation:-

$$nP_n(x) = xP'_n(x) - P'_{n-1}(x)$$

Solution:-

Consider L.D Polynomial

$$G(x,t) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n \quad \text{--- (i)}$$

Now

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x)t^n$$

Diff w.r.t x

$$-\frac{1}{2}(1-2xt+t^2)^{-3/2}(-2t) = \sum_{n=0}^{\infty} t^n P'_n(x)$$

$$= t(1-2xt+t^2)^{-3/2} = \sum_{n=0}^{\infty} t^n P'_n(x) \quad \text{--- (2)}$$

Now Diff eq (i)

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w.r.t 't'

$$= \frac{1}{2} (1 - 2xt + t^2)^{-3/2} (-2x + 2t) = \sum_{n=0}^{\infty} P_n(x) n t^{n-1}$$

$$(x-t) (1 - 2xt + t^2)^{-3/2} = \sum_{n=0}^{\infty} P_n(x) n t^{n-1} \quad \text{--- (3)}$$

Divide eq (3) by (2)

$$\frac{(x-t) (1 - 2xt + t^2)^{-3/2}}{t (1 - 2xt + t^2)^{-3/2}} = \frac{\sum_{n=0}^{\infty} P_n(x) n t^{n-1}}{\sum_{n=0}^{\infty} t^n P_n(x)}$$

$$(x-t) \sum_{n=0}^{\infty} t^n P'_n(x) = t \sum_{n=0}^{\infty} P_n(x) n t^{n-1}$$

$$x \sum_{n=0}^{\infty} t^n P'_n(x) - \sum_{n=0}^{\infty} t^{n+1} P'_n(x) = \sum_{n=0}^{\infty} P_n(x) n t^n$$

comparing coefficient of t^n

$$x P'_n(x) - P'_{n-1}(x) = n P_n(x)$$

$$n P_n(x) = x P'_n(x) - P'_{n-1}(x)$$

Proved:

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Q#4:-

The function $u_1(x)$ and $u_2(x)$ are eigenfunction of same hermitian operation but distinct eigen values λ_1 and λ_2 . Prove that $u_1(x)$ and $u_2(x)$ are linearly independent:-

Solution:-

Let $u_1, u_2, \dots, u_n$ be nonzero eigen vectors of A corresponding to distinct eigen values λ_1, λ_2 respectively.

We prove the theorem

by induction on n

if $n=1$ then u_1 is linearly

independent as $u_1 \neq 0$

suppose $n > 1$ and the result

is true for all $k < n-1$

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$$\text{Let } a_1 u_1 + a_2 u_2 + \dots + a_n u_n = 0$$

where $a_i \in \mathbb{R}$

$$\text{Then } A(a_1 u_1 + a_2 u_2 + \dots + a_n u_n) = A(0) = 0$$

$$a_1 A u_1 + a_2 A u_2 + \dots + a_n A u_n = 0$$

$$a_1 \lambda_1 u_1 + a_2 \lambda_2 u_2 + \dots + a_n \lambda_n u_n = 0 \quad \text{--- (2)}$$

Multiply by λ_n we get

$$a_1 \lambda_n u_1 + a_2 \lambda_n u_2 + \dots + a_n \lambda_n u_n = 0 \quad \text{--- (3)}$$

subtracting 3 from 2

$$a_1 (\lambda_1 - \lambda_n) u_1 + a_2 (\lambda_2 - \lambda_n) u_2 + \dots = 0$$

Since λ_1, λ_2 are all

distinct so $\lambda_1 - \lambda_n \neq 0$

For $i = 1, 2, \dots, n-1$

$$\text{Hence } a_1 = a_2 = \dots = a_{n-1} = 0$$

so eq. (i) becomes $a_n u_n = 0$

But $u_n \neq 0$ Therefore $a_n = 0$

Thus

u_1, u_2 are linearly independent.



2021

DATE: 28-Feb-23

DAY: Saturday

Kamran Qamar DIN

Roll no 40893

Assignment "MMP"

Past Paper "2021"

Group leader "Junaid"

Submitted to "H. Sir Hussain Farooq"

Short Q:-

Q2. "part 5"

Show that $P_n(x)$ are orthogonal and

$$\text{that: } \int_{-1}^1 P_n(x) P_m(x) dx = \frac{2}{2n+1}$$

when $m=n$.

Sol Case I

When $m=n$

Considering generating function of Legendre

$$(1-2xt+t^2)^{-1/2} = P_n(x) t^n \rightarrow (1)$$

$$(1-2xt+t^2)^{-1/2} = P_m(x) t^m \rightarrow (2)$$

Now, multiplying eq (1) and eq (2)

$$P_n(x) P_m(x) t^{n+m} = (1-2xt+t^2)^{-1/2} (1-2xt+t^2)^{-1/2}$$
$$= (1-2xt+t^2)^{-1}$$

Now integrating it from -1 to +1 then,

$$\int_{-1}^1 P_n(x) P_m(x) t^{n+m} dx = \int_{-1}^1 (1-2xt+t^2)^{-1} dx$$

$$= \frac{-1}{2t} \int_{-1}^1 \frac{-2t}{(1-2xt+t^2)} dx$$

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$$= \frac{-1}{2t} [\ln|1-2t+t^2| - \ln|1+2t+t^2|]$$

$$\int_{-1}^{+1} P_n(x) P_m(x) t^{m+n} dx = \frac{1}{2t} [\ln|1+2t+t^2| - \ln|1-2t+t^2|]$$

$$\begin{aligned} \int_{-1}^{+1} P_n(x) P_m(x) t^{m+n} dx &= \frac{1}{2t} [\ln|1+t|^2 - \ln|1-t|^2] \\ &= \frac{1}{t} \left[\frac{1}{2} \cdot 2 \ln|1+t| - \frac{1}{2} \cdot 2 \ln|1-t| \right] \\ &= \frac{1}{t} [\ln(1+t) - \ln(1-t)] \end{aligned}$$

$$\begin{aligned} \int_{-1}^{+1} P_n(x) P_m(x) t^{m+n} dx &= \frac{1}{t} \left[t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \frac{t^5}{5} - \dots \right. \\ &\quad \left. + t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{4} + \frac{t^5}{5} + \dots \right] \end{aligned}$$

$$\begin{aligned} \int_{-1}^{+1} P_n(x) P_m(x) t^{m+n} dx &= \left(2t + \frac{2t^3}{3} + \frac{2t^5}{5} + \dots \right) \\ &= 2 \left(1 + \frac{t^2}{3} + \frac{t^4}{5} + \dots \right) \end{aligned}$$

$$\int_{-1}^{+1} |P_n(x)|^2 t^{2n} dx = 2 \sum_{t=0}^{\infty} \frac{t^{2n}}{2n+1}$$

Comparing coefficient of t^{2n} , So,

$$\boxed{\int_{-1}^{+1} |P_n(x)|^2 dx = 2 \quad \text{Proved}}$$

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Q3

Part 4

Solve the eq of Simple Harmonic oscillator by using Laplace transform:

$$m \frac{d^2 x(t)}{dt^2} + Kx(t) = 0,$$

with initial condition:

$$x(0) = x_0, \quad \left. \frac{dx(t)}{dt} \right|_{t=0} = 0$$

$$m \frac{d^2 x(t)}{dt^2} + Kx(t) = 0$$

Taking Laplace transform of

$$m \mathcal{L} \left\{ \frac{d^2 x(t)}{dt^2} + Kx(t) \right\} = 0$$

$$m \left[s^2 X(s) - sX(0) \right] + KX(s) = 0$$

Divide by m , we get

$$s^2 X(s) - sX(0) + \frac{K}{m} X(s) = 0$$

$$s^2 X(s) - sX_0 + \omega^2 X(s) = 0$$

Put $\omega^2 = K/m$

$X(0) = x_0$

$$X(s)(s^2 + \omega^2) = sX_0$$

$$X(s) = \frac{sX_0}{s^2 + \omega^2}$$

Taking:

Inverse Laplace transform,

$$\mathcal{L}^{-1} \{ X(s) \} = x(t) = \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + \omega^2} \right\} x_0 =$$

$$x(t) = x_0 \cos(\omega t)$$

Eq of displacement of SH.O is

$$\int_{-\infty}^{+\infty} \frac{x^{k+1} e^{-xz} dz}{(1-t)^{k+1} (1+z)^{k+1}} = \int_0^{\infty} e^{-xz} x^{k+m} \sum_{t=0}^{\infty} \sum_{m=0}^{\infty} L_m^{(x)} L_n^{(x)} t^m z^n dz$$

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$$\int_0^{\infty} x^{k+1} e^{-\alpha x} = \frac{(k+1)!}{(\alpha)^{k+2}} = \frac{(k+1)!}{\alpha^{k+2}}, \text{ using}$$

$$\frac{1}{(1-t)^{k+1} (1-z)^{k+1}} \times \frac{(k+1)!}{\alpha^{k+2}} = \int_0^{\infty} e^{-\alpha x} x^{k+1} \sum_{t=0}^{\infty} \sum_{m=0}^{\infty} L_m(x) L_m(x) t^m z^m dx$$

$$\alpha = \frac{1-tz}{(1-t)(1-z)} \Rightarrow \frac{1}{\alpha} = \frac{(1-t)(1-z)}{(1-tz)}$$

$$\frac{1}{\alpha^{k+2}} = \frac{(1-t)^{k+2} (1-z)^{k+2}}{(1-tz)^{k+2}}, \text{ thus}$$

$$\frac{(k+1)!}{\alpha^{k+2}} = \frac{(k+1)! (1-t)^{k+2} (1-z)^{k+2}}{(1-tz)^{k+2}}$$

using expression get

$$\frac{(k+1)!}{(1-t)^{k+1} (1-z)^{k+1} \alpha^{k+2}} = \frac{(k+1)! (1-t)^{k+2} (1-z)^{k+1}}{(1-tz)^{k+2} (1-t)^{k+1} (1-z)^{k+1}}$$

$$= \frac{(k+1)! (1-t)(1-z)}{(1-tz)^{k+2}} \Rightarrow \textcircled{A}$$

Using the maclaurin Series

Since

$$\frac{1}{(1-z)^{k+1}} = \frac{(k+r)!}{k! r!} (tz)^r$$

replace k by k+1

$$\frac{(k+1)! (1-t)(1-z)}{(1-tz)^{k+2}} = \frac{(k+1)! (1-t)(1-z) (k+1+r)! (tz)^r}{(k+1)! r!}$$

$$\frac{(1-t)(1-z) (1+r)! (tz)^r}{r!} = \int_0^{\infty} e^{-\alpha x} x^{k+1} L_m(x) t^m z^m dx$$

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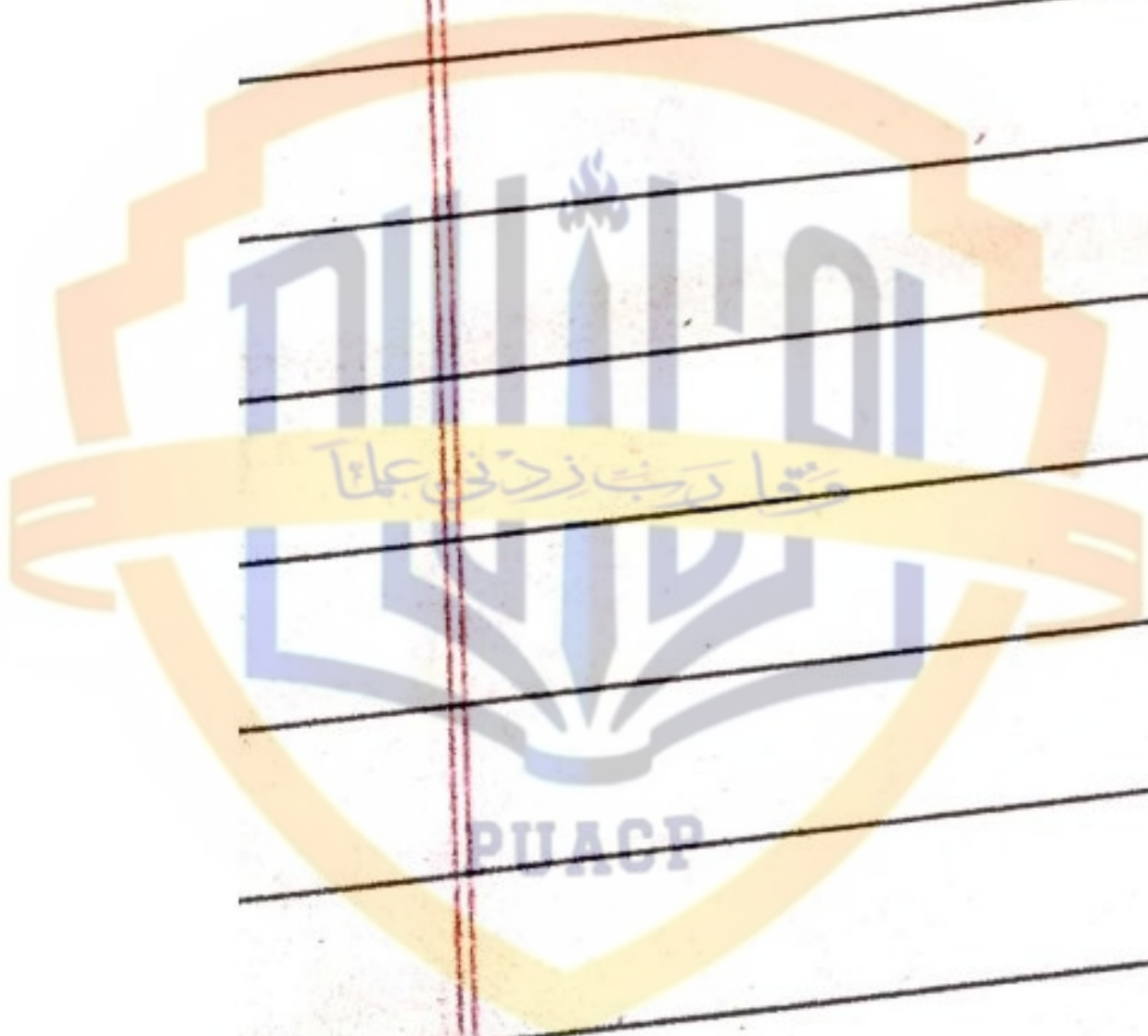
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Case I if $t=z$

$$\text{So, } \int_0^{\infty} e^{-x} x^{k+1} L_m(x) L_n(x) dx = \frac{(1-t)(1-z)(k+1)!}{k!} \rightarrow (1)$$

Case II if $t \neq z$ then

$$\int_0^{\infty} e^{-x} x^{k+1} L_m(x) L_n(x) dx = 0 \rightarrow (2)$$



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Q3 $\Rightarrow$

part 5

prove that: $|\Gamma(\alpha + i\beta)|$

$$= |\Gamma(\alpha)| \prod_{m=0}^{\infty} \left(1 + \frac{\beta^2}{(\alpha+m)^2}\right)^{-1/2}$$

The infinity Product of Gamma function is

$$\Gamma(z) = \frac{1}{z} \prod_{m=1}^{\infty} \left(1 + \frac{z}{m}\right)^{-1} n^{\frac{1}{n^2}} \quad \text{--- (A)}$$

Taking reciprocals,

$$\frac{1}{\Gamma(z)} = z \prod_{m=1}^{\infty} \left(1 + \frac{z}{m}\right)^{-1} n^{-\frac{1}{n^2}} \quad \text{--- (B)}$$

Put substitution $\Rightarrow n^{-\frac{1}{n^2}} = e^{-\ln(n)z}$

$$\frac{1}{\Gamma(z)} = z \prod_{m=1}^{\infty} \left(1 + \frac{z}{m}\right)^{-1} e^{-(\ln n)z}$$

Taking z^* , So (conjugate) of eq (B)

$$\frac{1}{\Gamma(z^*)} = z^* \prod_{m=1}^{\infty} \left(1 + \frac{z^*}{m}\right)^{-1} e^{-(\ln n)z^*} \quad \text{--- (C)}$$

multiply Eq (B) and (C)

$$\frac{1}{\Gamma(z)\Gamma(z^*)} = z z^* \prod_{m=1}^{\infty} \left(1 + \frac{z}{m}\right)^{-1} \left(1 + \frac{z^*}{m}\right)^{-1} e^{-(\ln n)(z+z^*)}$$

$$\text{let } z = \alpha + i\beta, \quad z^* = \alpha - i\beta$$

$$zz^* = \alpha^2 + \beta^2, \quad z+z^* = 2\alpha \quad \text{Now put}$$

$$\frac{1}{\Gamma(z)\Gamma(z^*)} = (\alpha^2 + \beta^2) \prod_{m=1}^{\infty} e^{-(\ln n)2\alpha \left(\frac{m+\alpha+i\beta}{m}\right) \left(\frac{m+\alpha-i\beta}{m}\right)}$$

$$\Gamma(\alpha+i\beta)\Gamma(\alpha-i\beta) = \alpha^2 \left(1 + \frac{\beta^2}{\alpha^2}\right) \prod_{m=1}^{\infty} e^{-(\ln n)2\alpha \left(\frac{m+\alpha+i\beta}{m}\right) \left(\frac{m+\alpha-i\beta}{m}\right)}$$

$$= \alpha^2 \left(1 + \frac{\beta^2}{\alpha^2}\right) \prod_{m=1}^{\infty} e^{-(\ln n)2\alpha \left(\frac{(m+\alpha)^2 + \beta^2}{m^2}\right)}$$

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$$= \alpha^2 \left(1 + \frac{\beta^2}{\alpha^2}\right) e^{-(\ln n)2\alpha} \prod_{m=1}^{\infty} \frac{(m+\alpha)^2}{m^2} \left(1 + \frac{\beta^2}{(m+\alpha)^2}\right)$$

$$= \alpha^2 \left(1 + \frac{\beta^2}{\alpha^2}\right) e^{-(\ln n)2\alpha} \prod_{m=1}^{\infty} \frac{m^2 \left(1 + \frac{\alpha}{m}\right)^2}{m^2} \left(1 + \frac{\beta^2}{(m+\alpha)^2}\right)$$

$$= \left(\alpha^2 \prod_{m=1}^{\infty} \left(1 + \frac{\alpha}{m}\right)^2 e^{-(\ln n)2\alpha} \right) \left(\frac{1+\beta^2}{\alpha^2} \prod_{m=1}^{\infty} \left(1 + \frac{\beta^2}{(m+\alpha)^2}\right) \right)$$

$$\prod_{m=0}^{\infty} \left(1 + \frac{\beta^2}{(m+\alpha)^2}\right)$$

Put $\frac{1}{(\Gamma(\alpha))^2} = \left(1 + \frac{\alpha}{m}\right)^2 \alpha^2 \prod_{m=1}^{\infty} e^{-(\ln n)2\alpha}$

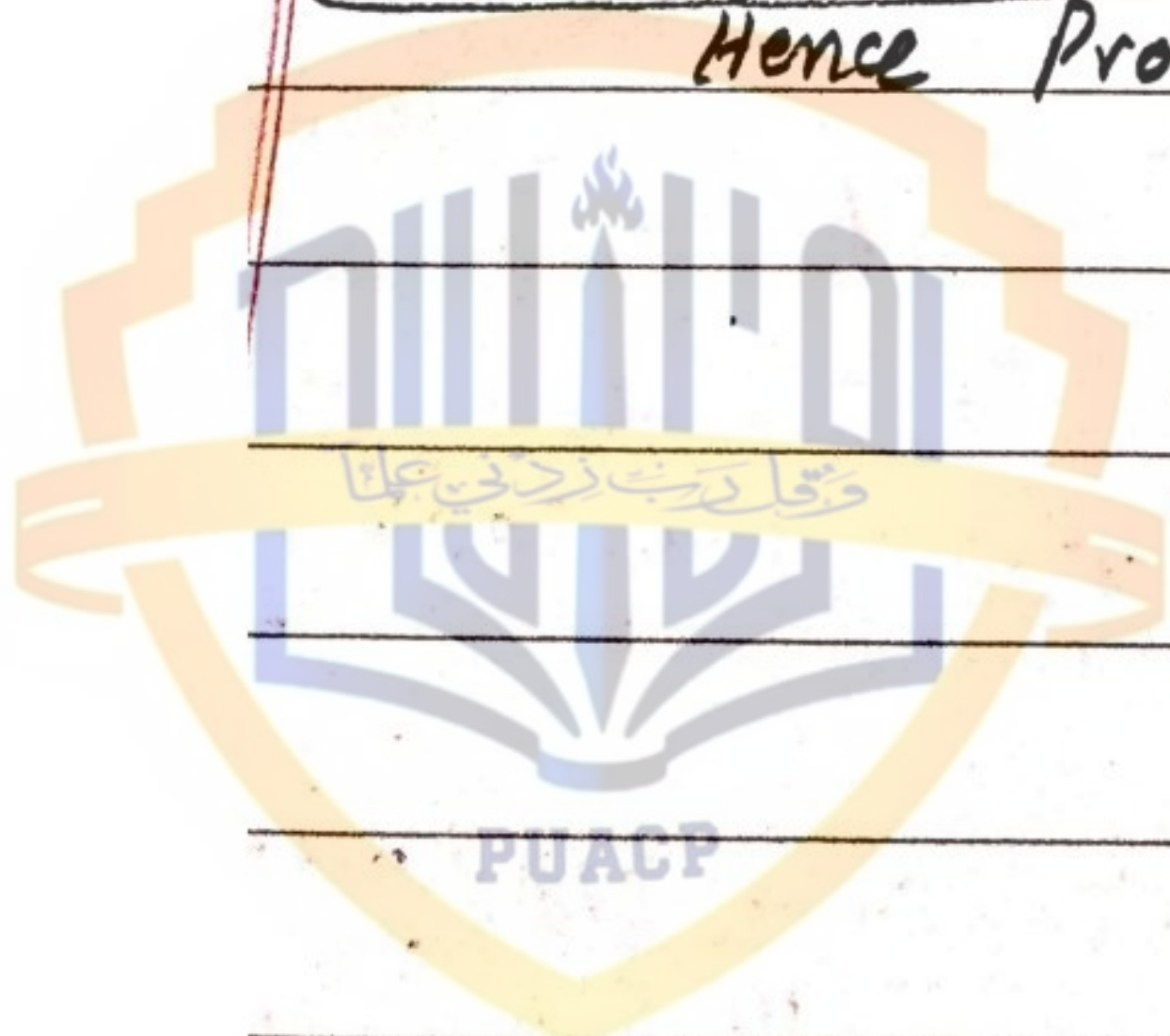
So we get.

$$\frac{1}{\Gamma(\alpha+i\beta)\Gamma(\alpha-i\beta)} = \frac{1}{|\Gamma(\alpha)|^2} \prod_{m=0}^{\infty} \left(1 + \frac{\beta^2}{(m+\alpha)^2}\right)$$

Taking reciprocal, square root
we get

$$\Gamma(\alpha+i\beta) = \Gamma(\alpha) \prod_{m=0}^{\infty} \left(1 + \frac{\beta^2}{(m+\alpha)^2}\right)^{-1/2}$$

Hence Proved.



→ Mathematical Method :-

→ Past paper ~ 2021 :-

Short Questions:-

1- Show that the recurrence relation:-

$$\frac{d}{dx} J_n(x) = \frac{2}{x} [J_{n-1}(x) - J_{n+1}(x)]$$

follows directly from substitution:-

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin\theta) d\theta.$$

Sol:-

$$\text{let } J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin\theta) d\theta.$$

$$\frac{d}{dx} J_n(x) = \frac{1}{\pi} \int_0^\pi -\sin(n\theta - x \sin\theta) \frac{d}{dx} (n\theta - x \sin\theta) d\theta$$

$$= \frac{1}{\pi} \int_0^\pi -\sin(n\theta - x \sin\theta) (-\sin\theta) d\theta$$

$$= \frac{1}{\pi} \int_0^\pi \sin(n\theta - x \sin\theta) \sin\theta d\theta.$$

As

$$-2 \sin \alpha \sin \beta = \cos(\alpha + \beta) - \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} (\cos(\alpha + \beta) - \cos(\alpha - \beta))$$

$$\text{put } \alpha = n\theta - x \sin\theta \text{ \& } \beta = \theta.$$

So

$$\frac{d}{dx} J_n(x) = \frac{1}{2\pi} \int_0^\pi \cos(n\theta - x \sin\theta - \theta) - \cos(n\theta - x \sin\theta + \theta) d\theta$$

$$= \frac{1}{2\pi} \int_0^\pi (\cos(n-1)\theta - x \sin\theta) d\theta - \frac{1}{2\pi} \int_0^\pi (\cos(n+1)\theta - x \sin\theta) d\theta$$

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Since:-

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin\theta) d\theta$$

$$J_{n-1}(x) = \frac{1}{\pi} \int_0^\pi (\cos(n-1)\theta - x \sin\theta) d\theta$$

$$J_{n+1}(x) = \frac{1}{\pi} \int_0^\pi (\cos(n+1)\theta - x \sin\theta) d\theta$$

So

$$\boxed{\frac{d}{dx} J_n(x) = \frac{1}{2} J_{n-1}(x) - \frac{1}{2} J_{n+1}(x)} \text{ Proved.}$$

2- Show that the Fourier sine and cosine transforms of e^{-at} are:-

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega}{\omega^2 + a^2}$$

$$g_c(\omega) = \sqrt{\frac{2}{\pi}} \frac{a}{\omega^2 + a^2}$$

Sol:-

For Fourier transform of sine is

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(\omega t) dt = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-at} \sin(\omega t) dt$$

By using formula:-

$$\int_0^\infty e^{-at} \sin bx dx = \frac{e^{-ax}}{a^2 + b^2} [a \sin bx - b \cos bx]_0^\infty$$

here case

$$a = -a \quad \text{and} \quad b = \omega$$

Thus-

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-at} \sin(\omega t) dt$$

$$= \sqrt{\frac{2}{\pi}} \frac{e^{-ax}}{(a^2 + \omega^2)} [-a \sin(\omega t) - b \cos(\omega t)]_0^\infty$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} \left(\frac{e^{-a\infty}}{a^2+b^2} [-a \sin(\infty) - b \cos(\infty)] \right)$$

$$- \left(\frac{e^0}{a^2+b^2} [-a \sin(0) - b \cos(0)] \right)$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} \left(0 - \frac{1}{a^2+b^2} (-a \cdot 0 - b(1)) \right)$$

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2+\omega^2}$$

Fourier cosine transformation:-

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-at} \cos(\omega t) dt = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos(\omega t) dt$$

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-at} \cos(\omega t) dt$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-at}}{a^2+\omega^2} (a \cos(\omega t) + b \sin(\omega t)) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[\left(\frac{e^{-\infty}}{a^2+\omega^2} (a \cos(\infty) + b \sin(\infty)) \right) - \left(\frac{e^0}{a^2+\omega^2} (a \cos(0) + b \sin(0)) \right) \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[0 + \frac{1}{a^2+\omega^2} (a) \right]$$

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \cdot \frac{a}{a^2+\omega^2}$$



$$= \frac{\sqrt{2}}{\sqrt{\pi}} \left(\frac{e^{-a\infty}}{a^2+b^2} [-a \sin(\infty) - b \cos(\infty)] \right)$$

$$- \left(\frac{e^0}{a^2+b^2} [-a \sin(0) - b \cos(0)] \right)$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} \left(0 - \frac{1}{a^2+b^2} (-a \cdot 0 - b(1)) \right)$$

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2+\omega^2}$$

Fourier cosine transformation:-

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-at} \cos(\omega t) dt = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos(\omega x) dx$$

So

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-at} \cos(\omega t) dt$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-at}}{a^2+\omega^2} (a \cos(\omega t) + b \sin(\omega t)) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[\left(\frac{e^{-\infty}}{a^2+\omega^2} (a \cos(\infty) + b \sin(\infty)) \right) - \left(\frac{e^0}{a^2+\omega^2} (a \cos(0) + b \sin(0)) \right) \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[0 + \frac{1}{a^2+\omega^2} (a) \right]$$

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \cdot \frac{a}{a^2+\omega^2}$$

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3- Expand $\delta(x-t)$ in Fourier series:-

$$\text{As } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} 1 \cdot \delta(x-t) dt$$

$$\text{Since:- } \int_{-\pi}^{\pi} f(x) \delta(x-t) = f(t)$$

If $f(x) = 1$ Thus:-

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} 1 \cdot \delta(x-t) dt$$

$$\Rightarrow a_0 = \frac{1}{\pi} \text{ --- (1)}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \delta(x-t) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} 1 \cdot \delta(x-t) \cos(nx) dx$$

$$\Rightarrow a_n = \frac{1}{\pi} \cos(nt) \text{ --- (2)}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \delta(x-t) \sin(nx) dx \quad \because f(x) = 1$$

$$\Rightarrow b_n = \frac{1}{\pi} \sin(nt) \text{ --- (3)}$$

So, Fourier series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

from eqn (2) and (3)

$$f(x) = \frac{1}{2\pi} + \sum_{n=1}^{\infty} \left(\frac{1}{\pi} \cos(nt) \cos(nx) + \frac{1}{\pi} \sin(nt) \sin(nx) \right)$$

$$f(x) = \frac{1}{2\pi} + \sum_{n=1}^{\infty} \frac{1}{\pi} \cos(nx - nt)$$

4. Show that $P_n(x)$ are orthogonal and that:-

$$\int_{-1}^{+1} P_n(x) P_n(x) dx = \frac{2}{2n+1}$$

Case-I ($m=n$)

Legendre function is

$$(1-2xt+t^2)^{-1/2} = P_n(x)t^n \quad \text{--- (1)}$$

$$(1-2xt+t^2)^{-1/2} = P_m(x)t^m \quad \text{--- (2)}$$

Multiply eqn (1) & (2)

$$P_n(x)t^n P_m(x)t^m = (1-2xt+t^2)^{-1/2} (1-2xt+t^2)^{-1/2}$$

$$= (1-2xt+t^2)^{-1}$$

Now integral from $-1 \rightarrow +1$

$$\int_{-1}^{+1} P_n(x)t^n P_m(x)t^m dx = \int_{-1}^{+1} (1-2xt+t^2)^{-1} dx$$

$$= \int_{-1}^{+1} \frac{-2t}{-2t(1-2xt+t^2)} dx$$

$$= -\frac{1}{2t} \left[\ln |1-2xt+t^2| \right]_{-1}^{+1}$$

$$= -\frac{1}{2t} \left[\ln |1-2t+t^2| - \ln |1+2t+t^2| \right]$$

$$= \frac{1}{2t} \left[\ln |1+2t+t^2| - \ln |1-2t+t^2| \right]$$

$$\int_{-1}^{+1} P_n(x) P_m(x) t^{n+m} dx = \frac{1}{2t} \left[\ln |1+t|^2 - \ln |1-t|^2 \right]$$

$$= \frac{1}{t} \left[\frac{1}{2} \cdot 2 \ln|1+t| - \frac{1}{2} \cdot 2 \ln|1-t| \right]$$

$$= \frac{1}{t} \left[\ln|1+t| - \ln|1-t| \right]$$

As $\ln|1+t| = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$

$$\ln|1-t| = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \dots\right)$$

So

$$= \frac{1}{t} \left(\left(t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots \right) + \left(t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{4} + \dots \right) \right)$$

$$= \frac{1}{t} \left(2t + \frac{2t^3}{3} + \frac{2t^5}{5} + \dots \right)$$

$$= 2 \left(1 + \frac{t^2}{3} + \frac{t^4}{5} + \dots \right)$$

$$\int_{-1}^{+1} |P_n(x)|^2 t^{2n} dx = 2 \sum_{n=0}^{\infty} \frac{t^{2n}}{2n+1} \quad \because m=n$$

Comparing

$$\int_{-1}^{+1} |P_n(x)|^2 dx = \frac{2}{2n+1}$$

Proved

Long Questions:-

Construct a green function:- for:-

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (k^2 x^2 - 1)y = 0$$

Boundary condition $y(0)=0, y(1)=0$

Sol:- It's the Bessel eqn in form of

$$x^2 y'' + xy' + (a^2 x^2 - n^2)y = 0$$

Now comparing $a^2 = k^2 \Rightarrow a = \pm k$

$$n^2 = 1 \Rightarrow n = \pm 1$$

So $y = C_1 J_n(ax) + C_2 J_{-n}(ax)$

$$y = C_1 J_1(kx) + C_2 J_{-1}(kx)$$

the operator of $\mathcal{L}$ of diff eqn

$$P(x) D^2 + q D + r = \mathcal{L}$$

$$x^2 \frac{d^2}{dx^2} + x \frac{d}{dx} + (k^2 x^2 - 1) = \mathcal{L}$$

and this operator on $G(x, x')$, then

$$\mathcal{L} G(x, x') = \delta(x - x')$$

when $x \neq x'$

$$\delta(x - x') = 0, \text{ so,}$$

$$\mathcal{L} G(x, x') = 0$$

So

$$\left(x^2 \frac{d^2}{dx^2} + x \frac{d}{dx} + (k^2 x^2 - 1) \right) G(x, x') = 0$$

The Bessel eqn roots are:

$$J_1(kx) \text{ and } J_0(kx)$$

So general sol is

$$y = G(x, x') = C_1 J_1(kx) + C_2 J_0(kx)$$

⇒ When $x < x'$, the Green function is left Green function, so apply boundary.

$$\Rightarrow y(0) = 0$$

$$\text{So } G(0, x') = 0$$

$$C_1 J_1(k(0)) + C_2 J_0(k(0)) = 0$$

$$C_1 J_1(0) + C_2 J_0(0) = 0$$

$$C_1(0) + C_2(0) = 0$$

We cannot find the value of C_1 & C_2 , so

$$G(x, x') = C_1 J_1(kx) + C_2 J_0(kx)$$

$$\Rightarrow C_1 = C \text{ and } C_2 = D$$

$$G(x, x') = C J_1(kx) + D J_0(kx)$$

⇒ When $x > x'$

$$G(x, x') = C_1 J_1(k) + C_2 J_0(k) = 0$$

$$C_1 J_1(k) = -C_2 J_0(k)$$

$$C_1 = \frac{-C_2 J_0(k)}{J_1(k)}$$

put value of C_1 this.

$$G_2(1, x') = - \frac{C_2 J_{-1}(k)}{J_1(k)} J_1(kx) + C_2 J_{-1}(kx)$$

$$= \frac{C_2}{J_1(k)} \left(J_1(kx) - J_{-1}(k) J_1(kx) \right)$$

Put $C_2 = 0$

Now apply eqn of continuity,

$$G_2(x, x') = G_1(x, x') \text{ thus,}$$

$$C_2 J_1(kx) + D J_{-1}(kx) = -D \frac{J_{-1}(k)}{J_1(k)} J_1(kx)$$

$$D J_{-1}(kx) = -D \frac{J_{-1}(k)}{J_1(k)} \quad \text{--- (1)}$$

Now

$$P'(x) \left\{ \frac{G_2(x, x')}{dx'} \bigg|_{x=x'} - \frac{G_1(x, x')}{dx} \bigg|_{x=x'} \right\} = -1$$

on comparing eqn $\boxed{P(x') = x'^2}$

$$\frac{dG_2(x, x')}{dx} \bigg|_{x=x'} = C J_1'(kx') + D J_{-1}'(kx')$$

$$\frac{dG_1(x, x')}{dx} \bigg|_{x=x'} = D J_{-1}'(kx') - \frac{D J_{-1}(k)}{J_1(k)} J_1'(kx')$$

Now apply values

$$C J_1'(kx') + D J_{-1}'(kx') - D J_{-1}'(kx') + \frac{D J_{-1}(k)}{J_1(k)} J_1'(kx') = \frac{-1}{x'^2}$$

- Eqn of S.H.O by using Laplace transform.

$$m \frac{d^2 x(t)}{dt^2} + k x(t) = 0,$$

with initial condition

$$x(0) = x_0, \quad \left. \frac{dx(t)}{dt} \right|_{t=0} = 0$$

Sol:- $m \frac{d^2 x(t)}{dt^2} + k x(t) = 0$

Taking L-T

$$m \mathcal{L} \left\{ \frac{d^2 x(t)}{dt^2} + k x(t) \right\} = 0$$

$$m (s^2 x(s) - s x(0)) + k x(s) = 0$$

Divide by m

$$s^2 x(s) - s x(0) + \frac{k}{m} x(s) = 0$$

$$s^2 x(s) - s x_0 + \frac{k}{m} x(s) = 0$$

$$x(s) (s^2 + \omega^2) = + s x_0$$

$$x(s) = \frac{s x_0}{s^2 + \omega^2}$$

Taking inverse L-T

$$\mathcal{L}^{-1} \{ x(s) \} = x(t) = \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + \omega^2} \right\} x_0 =$$

$$x(t) = x_0 \cos(\omega t)$$

eqn of displacement of S.H.O

$$\int_{-\infty}^{+\infty} \frac{x^{k+1} e^{-\alpha x}}{(1-t)(1+z)^{k+1}} dx = \int_{-\infty}^{+\infty} e^{-x} x^{k+m} \sum_{t=0}^{\infty} \sum_{m=0}^{\infty} \frac{(x)^k}{m!} \frac{(1)^m}{n!} t^m z^n dx$$

Using expression $\int_0^{\infty} x^{k+1} e^{-\alpha x} dx = \frac{(k+1)!}{(\alpha)^{k+2}} = \frac{(k+1)!}{(1-tz)^{k+2}}$

$$\frac{1}{(1-t)^{k+1} (1-z)^{k+1}} \frac{(k+1)!}{k+2} = \int_0^{\infty} e^{-x} x^{k+1} \sum_{t=0}^{\infty} \sum_{m=0}^{\infty} \frac{(x)^k}{m!} \frac{(1)^m}{n!} t^m z^n dx$$

$$\alpha = \frac{1-tz}{(1-t)(1-z)} \Rightarrow \frac{1}{\alpha} = \frac{(1-t)(1-z)}{1-tz}$$

$$\frac{1}{\alpha^{k+2}} = \frac{(1-t)^{k+2} (1-z)^{k+2}}{(1-tz)^{k+2}} \text{ thus}$$

$$\frac{(k+1)!}{\alpha^{k+2}} = \frac{(k+1)!}{(1-tz)^{k+2}} \cdot \frac{(1-t)^{k+2} (1-z)^{k+2}}{(1-tz)^{k+2}}$$

$$\frac{(k+1)!}{(1-t)^{k+1} (1-z)^{k+1} \alpha^{k+2}} = \frac{(k+1)! (1-t)(1-z)}{(1-tz)^{k+2}} \quad \text{A}$$

Using Maclaurin series:

$$\frac{(k+1)! (1-t)(1-z)}{(1-tz)^{k+2}} = \frac{(k+1)! (1-t)(1-z) (k+1+1)! (kz)^1}{(k+1)! 1! \cdot 1}$$

$$\frac{(1-t)(1-z) (k+1+1)! (kz)^1}{1!} = \int_0^{\infty} e^{-x} x^{k+1} \frac{(x)^k}{m!} \frac{(1)^m}{n!} t^m z^n dx$$

→ Case-I if $t = z$

So, $\int_0^\infty e^{-x} x^{k+1} L_m(x) L_n(x) dx = \frac{(k+1)(1-z)(k+1+\delta)!}{k!} \quad (1)$

→ Case-2 If $t \neq z$

$\int_0^\infty e^{-x} x^{k+1} L_m(x) L_n(x) dx = 0 \rightarrow (2)$

$x \text{ ————— } x$

